

**A Statistical Information Theory of Evaporating Black Holes: Discrete Packet Emissions,
Continuous Diffusions, and the Information Cost of Coarse-Graining**

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Abstract

This report develops a measure-theoretic and computational framework for testing whether a smooth, continuum description of black-hole evaporation suppresses time-resolved information signatures that remain visible under discrete quantum emission. Paradigm A represents mass loss as a stopped, non-homogeneous marked point process with intensity γ/M and Planck-distributed energy marks. Paradigm B represents mass loss as a stopped Itô diffusion with drift $-\alpha/M^2$ and state-dependent volatility σ_0/M . Both are embedded in a common filtration and detector-operator framework so that latent dynamics can be separated from observation design. Exact complete-path likelihoods, a Girsanov likelihood, entropy-drift identities, score-based Fisher information, common-coordinate pullbacks, and continuous-time information functionals are derived. A reproducible PyTorch implementation uses fixed-data autograd scores and pathwise sensitivities while stopping trajectories above a positive mass boundary. In the validated baseline (16 replicates), the packet model produced an open-loop information diagnostic of 9.6041×10^{-8} nats and the requested diffusion 1.9774×10^{-8} nats, giving $\Delta I = -7.6267 \times 10^{-8}$ nats. This direction is consistent with—but does not establish—the coarse-graining hypothesis because the current reduced predictor is open loop, the latent laws are not fully moment matched, and Monte Carlo uncertainty is large. The decisive study therefore requires a common detector, causal observation-history filters, moment matching, uncertainty quantification, cutoff refinement, and simulation-based calibration.

Keywords: black-hole evaporation, marked point process, stochastic differential equation, transfer entropy, Fisher information, PyTorch

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Research Problem and Inferential Target

Black-hole evaporation is often summarized by equilibrium temperature, entropy, and mean power even though it is time ordered and terminating. Hawking radiation has temperature inversely proportional to mass, while Bekenstein's area law assigns the horizon a mass-dependent entropy (Bekenstein, 1973; Hawking, 1975). The information problem arises because a thermal output appears unable to encode the initial state, whereas unitarity requires total quantum evolution to preserve information (Harlow, 2016; Page, 1993). The radiation record should therefore be treated as a stochastic experiment rather than equilibrium snapshots.

The directional hypothesis is that, after controlling local moments, detector, horizon, stopping rule, and estimand, a continuous diffusion contains less directed information about latent mass than an event-resolving packet process. Let $\mathcal{E}^A$ and $\mathcal{E}^B$ denote the experiments and $I^A(T)$, $I^B(T)$ their filtration-valid information over $[0, T]$. The target $\Delta I(T) = I^B(T) - I^A(T)$ supports masking only when it is persistently negative with uncertainty excluding zero across pre-specified robustness analyses. One negative plug-in estimate is not proof because unmatched laws, unequal sensors, biased filters, or Monte Carlo noise can produce the same sign.

Contributions

The framework contributes a stopped Planck-marked process, a stopped diffusion with Girsanov likelihood, a common observation filtration, distinct statistical and pathwise derivatives, and a PyTorch architecture for gradient-based inference. The baseline findings are used to define the next confirmatory protocol.

Physical and Thermodynamic Basis

Natural Planck units $G = c = \hbar = k_B = 1$ are used. For a nonrotating, uncharged Schwarzschild black hole, the Hawking temperature and Bekenstein–Hawking entropy reduce to functions of mass alone.

$$T_h(M) = \kappa_h/M, \quad \kappa_h = 1/(8\pi) \quad (1)$$

$$S^{BH}(M) = A/4 = 4\pi M^2 \quad (2)$$

These relations determine the singular structure of the statistical models. As M decreases, temperature, emission intensity, drift magnitude, and process volatility increase. Neither paradigm is therefore stationary. The semiclassical formulas are not trusted at $M = 0$, so both processes are stopped at a positive boundary m_s . For $P \in \{A, B\}$, define $\tau_s^P = \inf\{t \geq 0 : M_t^P \leq m_s\}$ and analyze $M^P(t \wedge \tau_s^P)$. The cutoff is both a physical statement about the model's domain and a regularization required for stable likelihood and gradient calculations.

If a deterministic mean law obeys $dM/dt = -\alpha/M^2$, then $M(t)^3 = M_0^3 - 3\alpha t$ and the nominal lifetime is $\tau_0 = M_0^3/(3\alpha)$. This cubic relation creates an inferential ridge between initial mass and drift. It also shows why end-stage observations dominate curvature: small mass changes imply large changes in local temperature and flux. The resulting apparent information gain can be numerically genuine but physically outside the valid semiclassical regime. Every terminal claim must therefore be reported as a function of m_s , the time step, and the assumed greybody/mark law.

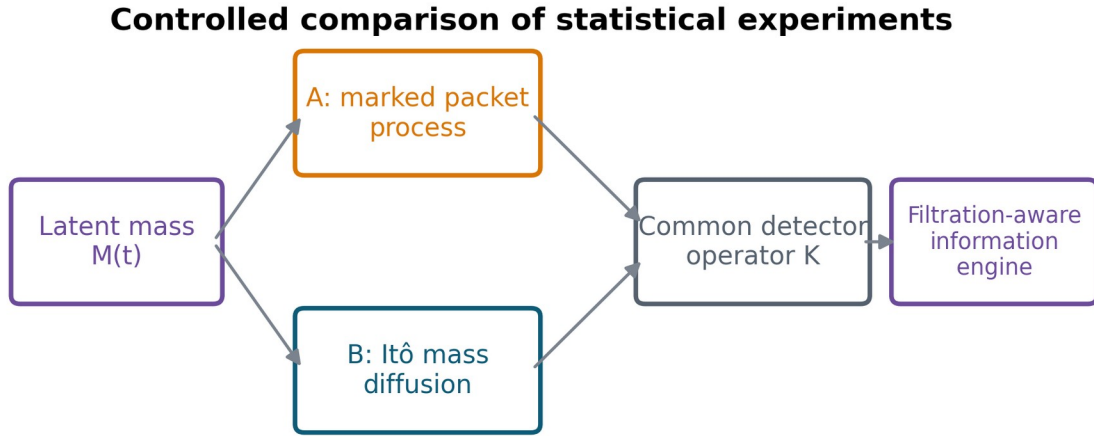
The proposed comparison concerns statistical representations of a semiclassical process; it is not itself a microscopic theory of quantum gravity. Discrete packet timing can preserve temporal structure without proving that spacetime is fundamentally discrete. The estimand must remain tied to observable experiment design.

Unified Statistical Experiment

Let $(\Omega, \mathcal{F}, \{\mathcal{F}(t): 0 \leq t \leq T\}, P_\theta)$ satisfy the usual conditions, with $\theta = (M_0, \gamma, \alpha)$. Each paradigm generates a latent mass process M^P adapted to a latent filtration $\mathcal{F}^{M,P}(t)$. A detector operator K maps the latent path or its emitted radiation to telemetry Y^P , whose observation filtration is $\mathcal{F}^{Y,P}(t) = \sigma\{Y^P(s): s \leq t\}$. The analysis filtration is $\mathcal{F}^P(t) = \mathcal{F}^{M,P}(t) \vee \mathcal{F}^{Y,P}(t)$, augmented by P_θ -null sets. This construction permits point-process and diffusion likelihoods on the same horizon while keeping their natural sample spaces.

Figure 1

Controlled Comparison of Statistical Experiments



Note. The detector operator must be identical across paradigms for the primary causal comparison. The current baseline uses paradigm-specific open-loop predictors and is therefore diagnostic only.

A clean experiment compares the observation laws induced by applying the same K to each latent law. Without a common K , a difference in information may be caused by the sensor rather than the latent law. This distinction motivates a 2×2 control design: jump versus diffusion latent dynamics crossed with event-resolving versus continuous telemetry. The interaction isolates the cost of smoothing the dynamics from the cost of smoothing the detector.

Paradigm A: Discrete Packet Jump Process

Paradigm A is a non-homogeneous marked point process. Let $N(dt, de)$ be an integer-valued random measure on $[0, T] \times (0, \infty)$ with predictable compensator $v\theta(dt, de) = 1_{\{t < \tau_s^A\}} \lambda(M_{t-}^A) f_e(e|M_{t-}^A) de dt$. The state equation and intensity are

$$dM_t^A = -\int_0^\infty e N(dt, de), \quad \lambda(M) = \gamma/M \quad (3)$$

so a realized packet E_i produces $M(t_i) = M(t_i-) - E_i$. Overshooting marks are censored or mapped to the stopping atom at m_s ; treating an overshoot as an ordinary density contribution would miss the terminal probability mass. The rate diverges as $M \rightarrow 0$, implying event clustering near the endpoint.

Table 1

Comparison of the Two Latent Laws

Feature	Paradigm A	Paradigm B
Latent law	Stopped marked point process	Stopped Itô diffusion
Mass change	Discrete energy jumps	Continuous Euler-Maruyama increments
Local driver	$\lambda(M)=\gamma/M$	$b(M)=-\alpha/M^2$; $s(M)=\sigma_o/M$
Natural data	Event times and energy marks	Continuous noisy telemetry
Inactive parameter	α	γ

Note. Paradigm-specific inactive parameters are structural, not numerical, zeros in the native Fisher information matrix.

This representation retains event topology: the count, ordered times, and packet energies. It is thus a strictly richer raw record than an integrated flux if the detector can resolve individual packets. Whether that richness survives a common sensor is an empirical question, not an assumption.

Thermal Marks and Point-Process Likelihood

Write $E = T_h(M)X$ and use the bosonic number-spectrum density $f_x(x) = x^2/[(e^x-1)2\zeta(3)]$, $x>0$. Its mean and second moment are $c1 = \pi^4/[30\zeta(3)] = 2.701195$ and $c2 = 12\zeta(5)/\zeta(3) = 10.351600$. Direct inverse-CDF sampling is avoided through the expansion $(e^x-1)^{-1} = \sum_{k \geq 1} e^{-kx}$. Consequently, K has probability proportional to k^{-3} , and $X|K=k$ follows Gamma(shape=3, rate=k). A finite mixture truncation is differentiable with respect to the temperature scale once the categorical and Gamma base randomness is held fixed.

For observed event times t_i and marks e_i before the boundary, the complete-path log likelihood is

$$\ell_a(\theta) = \sum_i [\log \lambda(M_{i-}) + \log f_e(e_i|M_{i-})] - \int_0^T \lambda(M_t) dt + \ell_0 \quad (4)$$

where ℓ_0 is the initial calibration likelihood. The compensator is essential: a likelihood based only on event contributions rewards arbitrarily high intensity. If the final packet crosses m_s , the last density is replaced by $\log P(E \geq M_{t-} - m_s | M_{t-})$, preserving the terminal atom.

Differentiable Event Times

A unit exponential variable Z supplies the next waiting time through the time-rescaling equation $\int_{t-}^{t+\Delta} \lambda(M_s) ds = Z$. Between jumps M is constant in the baseline, giving $\Delta = ZM/\gamma$. This reparameterizes waiting times and allows pathwise gradients almost surely, although the derivative is undefined on the measure-zero surfaces where the event count changes. Statistical scores must therefore be computed with realized data fixed, while pathwise derivatives answer a separate design-sensitivity question.

Paradigm B: Continuous Quantum-Field Flow

Paradigm B replaces individual emissions with a continuous stochastic mass flux. Before stopping,

$$dM_t^B = b(M_t^B; \alpha)dt + s(M_t^B)dW_t, \quad b(M; \alpha) = -\alpha/M^2, \quad s(M) = \sigma_0/M \quad (5)$$

where W is a Brownian motion under P_θ . Euler–Maruyama on a regular grid $t_j = j\Delta t$ gives

$$M_{j+1} = M_j - (\alpha/M_j^2)\Delta t + (\sigma_0/M_j)\sqrt{\Delta t} \varepsilon_j, \quad \varepsilon_j \sim N(0,1) \quad (6)$$

with an absorbing update at m_s . The transition log likelihood conditional on the discretized path is a sum of Gaussian terms with mean $M_j - \alpha\Delta t/M_j^2$ and variance $\sigma_0^2\Delta t/M_j^2$, plus the initial calibration term. Because σ_0/M does not depend on α , the drift likelihood relative to the zero-drift reference with the same diffusion coefficient follows from Girsanov's theorem:

$$\log(dP_\beta/dP_0) = \int_0^T \tau_s^B (b/s^2) dM_t - (1/2) \int_0^T \tau_s^B (b^2/s^2) dt \quad (7)$$

Absolute continuity requires the stopped process and integrability conditions such as Novikov's criterion. The positive boundary prevents the ratios b/s and b/s^2 from diverging on the analyzed interval. The formula is not valid if the reference process changes the diffusion coefficient.

Entropy Drift

Applying Itô's lemma to $S^{BH}(M) = 4\pi M^2$ yields an expected local entropy drift $8\pi M b(M) + 4\pi \tau_s(M)^2 = -8\pi\alpha/M + 4\pi\sigma_0^2/M^2$. The second term is an Itô correction; omitting it biases the continuous entropy budget, particularly near the cutoff.

Stopping, Stability, and the Final-Mass Singularity

The requested diffusion contains two increasingly stiff local scales. Requiring the deterministic decrement to be small relative to M and the stochastic standard deviation to be small relative to M gives

$$\Delta t \ll M^3/\alpha, \quad \Delta t \ll M^4/\sigma_0^2 \quad (8)$$

so a fixed grid that is adequate at M_0 can fail near m_s . Adaptive stepping based only on state magnitude may also bias a likelihood if the adaptation rule is not incorporated in the observation scheme. Recommended controls are deterministic refinement schedules, event-driven boundary localization, and convergence plots over $(\Delta t, m_s)$.

The packet model has a different terminal pathology. Its intensity y/M rises, mark temperature κ_h/M rises, and the quadratic variation rate rises faster still. Simulation must distinguish a genuine high-rate cluster from an algorithmic cascade caused by a stale state or truncated waiting time. The implementation therefore uses positive parameter transforms, a maximum-event guard, a terminal survival contribution, and explicit provenance for the stopping rule.

Boundary-Induced Information

Likelihood curvature and Fisher information can grow near the endpoint without implying unlimited physical information. The hard boundary encodes where semiclassical dynamics are declared invalid. Information peaks must be decomposed into interior and stopping-event contributions, making sensitivity to m_s a primary analysis.

For HMC/NUTS, boundary proximity produces large gradients and divergent transitions. Softplus-transformed raw parameters, noncentered Brownian innovations, and separated detector/process noise improve—but do not cure—geometry.

Moment Matching and the Coarse-Graining Null

The local conditional mean and quadratic-variation rates of Paradigm A follow directly from the compensator:

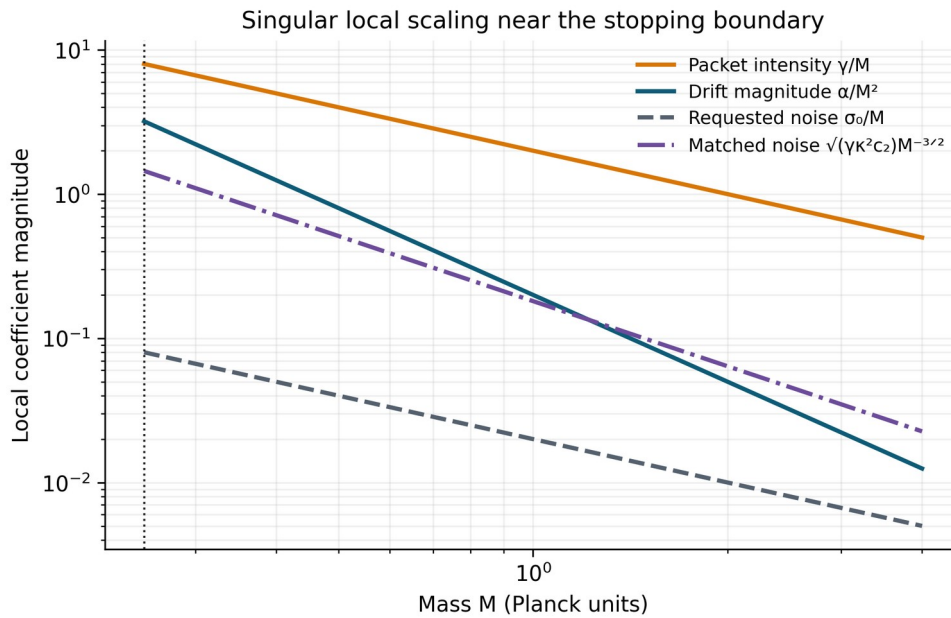
$$E[dM_t^A | \mathcal{F}_{t-}] / dt = -\gamma \kappa_h c_1 / M_t^2 \quad (9)$$

$$d\langle M^A \rangle_t / dt = \gamma \kappa_h^2 c_2 / M_t^3 \quad (10)$$

Drift matching sets $\alpha = \gamma \kappa_h c_1$. Variance matching requires volatility proportional to $M^{-3/2}$; the requested σ_0/M instead scales as M^{-1} and cannot be matched globally with one σ_0 . The baseline matched α is 0.214954 versus the requested 0.200000, a difference of -0.014954 .

Figure 2

Singular Local Scaling Near the Stopping Boundary



Note. The different volatility exponents mean the requested A-versus-B comparison combines discretization and local-law mismatch.

The primary null should compare A with moment-matched B★ under the same detector.

Requested B remains a sensitivity model; reporting both contrasts separates smoothing from unequal conditional moments.

Observation Filtration and Continuous-Time Information

Transfer entropy is a likelihood-ratio functional between a full predictor conditioned on latent and observed history and a reduced predictor conditioned only on observed history (Schreiber, 2000; Spinney et al., 2017). For an event detector with predictable intensities λ_t^{full} and λ_t^{red} , the local rate is

$$T_{\text{event}}(t) = E[\lambda_t^{\text{full}} \log(\lambda_t^{\text{full}}/\lambda_t^{\text{red}}) - \lambda_t^{\text{full}} + \lambda_t^{\text{red}}] \quad (11)$$

For a Gaussian observation channel $dY_t = h(M_t)dt + \rho dV_t$, Duncan's identity gives a directed information rate of the form

$$T_{\text{Gauss}}(t) = (1/2\rho^2) E[(h(M_t) - E[h(M_t)|\mathcal{F}_t^Y])^2] \quad (12)$$

under the channel's regularity assumptions (Duncan, 1970). The reduced intensity or conditional mean must be generated by a causal filter adapted to $\mathcal{F}_t^Y$. Replacing it with an unconditional or leave-one-out population mean creates an open-loop likelihood-information contrast, not transfer entropy.

Required Detector Controls

The decisive experiment applies one detector operator K to both latent laws. A practical design can model finite quantum efficiency, time jitter, energy resolution, dead time, and additive telemetry noise. The filter must then be trained or integrated separately under each law but evaluated on identical detector settings. A 2×2 design—latent packet/diffusion crossed with event/continuous observation—tests whether any gap is due to latent smoothing, measurement smoothing, or their interaction.

Information estimates should be integrated over a common at-risk process $1_{\{t < \tau_s^A \wedge \tau_s^B\}}$ or reported with a carefully defined terminal contribution. Otherwise, differing stopping times create unequal exposure. Bootstrap or nested Monte Carlo intervals must accompany $\Delta I(t)$ curves and their integrated values.

Fisher Information and Derivative Semantics

For fixed observed data y , the score and Fisher information are $s_\theta(y) = \partial \ell(\theta; y) / \partial \theta$ and $J(\theta) = E_\theta[s_\theta s_\theta^T]$. The repository estimates J by the empirical outer product of fixed-data autograd scores. This differs from differentiating a simulated path $y = g(\theta, \epsilon)$, which returns a pathwise sensitivity containing both likelihood and data-generation derivatives. It also differs from derivatives across event-count changes, which are distributional rather than ordinary pathwise derivatives.

$$\mathcal{J}^R(\theta) = R^{-1} \sum_{r=1}^R s_r(\theta) s_r(\theta)^T \quad (13)$$

Native parameter vectors contain structural zeros: α is inactive in A and γ is inactive in B . Directly subtracting their 3×3 Fisher matrices would therefore compare different coordinate meanings. Let the common coordinate be $\psi = (M_o, \alpha)$, with $\alpha \approx \gamma \kappa_h c_1$ for A and $\alpha \approx \alpha$ for B . If $\theta^P = g^P(\psi)$, then

$$J(\psi; P) = G_p^T J(\theta; P) G_p, \quad G_p = \partial g_p(\psi) / \partial \psi \quad (14)$$

and $\Delta J \psi = J \psi^B - J \psi^A$ is interpretable as a local difference in precision for the same estimands. Positive semidefiniteness is checked numerically, but the residual need not be positive semidefinite because neither experiment is assumed to Blackwell-dominate the other.

Missing Information

Complete-path Fisher information exceeds observed-data information when latent states or packets are hidden. Louis's identity decomposes observed curvature into complete-data information minus missing information associated with the conditional latent distribution (Louis, 1982). This distinction is central: a large complete-path matrix does not imply a realizable detector can attain that precision.

PyTorch Architecture and Numerical Validation

The repository implements both processes in pure PyTorch with float64 arithmetic. Physical parameters are represented by unconstrained raw tensors and mapped through softplus transforms. Random draws are stored as base noise so that common-random-number path comparisons and pathwise derivatives are reproducible. The central engine runs both paradigms, evaluates complete-path and Girsanov likelihoods, computes fixed-data scores, pulls Fisher matrices into common coordinates, and records information and entropy diagnostics.

Table 2

Validated Baseline Configuration

Quantity	Symbol	Baseline
Initial mass	M_0	4.0
Packet-rate scale	γ	2.0
Diffusion drift	α	0.2
Time horizon	T	0.2
Euler step	Δt	0.01
Stopping mass	m_s	0.25
Diffusion scale	σ_0	0.02
Replicates	R	16

Note. All calculations use Planck units, a positive stopping boundary, and seed 20260820.

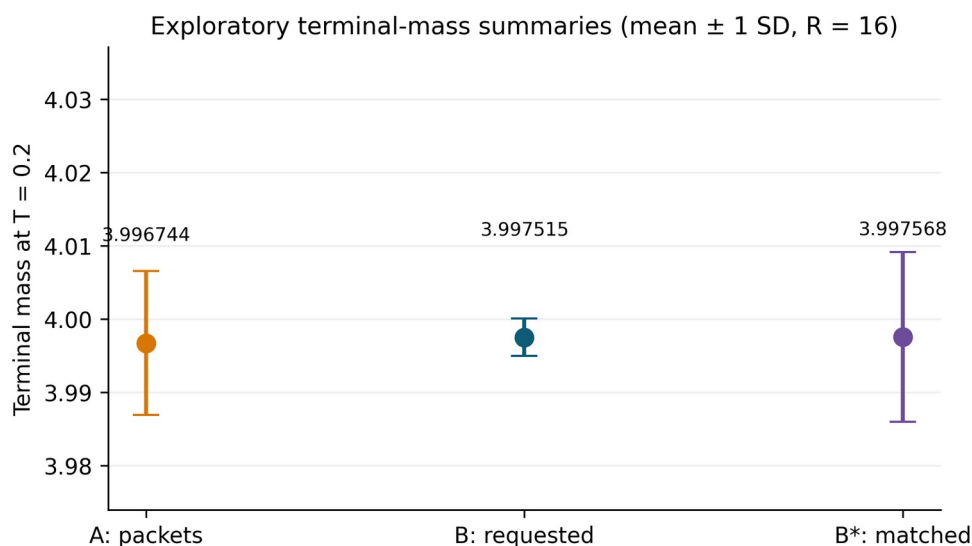
Six numerical tests passed in the saved validation environment: Planck moments; positivity and moment matching; packet likelihood and fixed-data score; diffusion likelihood, Girsanov agreement, and score; Fisher shapes and positive-semidefinite checks; and JSON serialization. The archived runtime reports PyTorch 2.13.0+cu130. A fresh rerun was not substituted for this provenance because the present document environment does not include PyTorch. The report therefore treats the saved artifacts as the validated computational record and states that dependency limitation explicitly.

Baseline Trajectory Findings

The exploratory baseline used $R = 16$ replicates over $T = 0.2$. Paradigm A produced a mean event count of 0.125 ($SD=0.331$), a mean terminal mass of 3.996744 ($SD=0.009799$), and no boundary hits. The requested diffusion produced a mean terminal mass of 3.997515 ($SD=0.002559$); the moment-matched $B^\star$ produced 3.997568 ($SD=0.011578$). Neither diffusion hit the boundary.

Figure 3

Exploratory Terminal-Mass Summaries



Note. Error bars show between-replicate standard deviations, not confidence intervals. $R = 16$ is too small for stable tail or endpoint inference.

Table 3

Baseline Information and Entropy Findings

Estimand	Value	Status
I^A	9.6041×10^{-8}	Open-loop diagnostic
I^B	1.9774×10^{-8}	Open-loop diagnostic
$\Delta I = I^B - I^A$	-7.6267×10^{-8}	Exploratory
Mean entropy drift A	-1.344414	Validated calculation
Mean entropy drift B	-1.256689	Validated calculation
Unit tests	6	All passed

The short horizon explains the similar means. $B^\star$ has variance closer to A than requested B, consistent with local quadratic-variation matching; none of these summaries estimates a physical endpoint.

Common-Coordinate Fisher Findings

In $\psi=(M_0, \tilde{\alpha})$ coordinates, the empirical Fisher matrices were

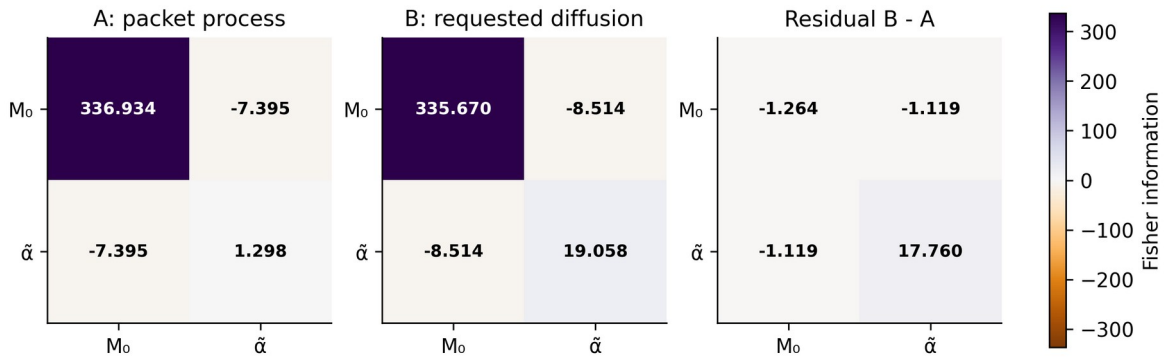
$$J^A = [[336.934, -7.395], [-7.395, 1.298]] \quad (15)$$

$$J^B = [[335.670, -8.514], [-8.514, 19.058]] \quad (16)$$

so $\Delta J=B-A$ had diagonal entries -1.264 for M_0 and $+17.760$ for $\tilde{\alpha}$, with off-diagonal entries -1.119 .

Figure 4

Empirical Fisher Information in Common Coordinates



Note. Matrices are empirical outer products from 16 fixed-data scores. Their numerical scale should not be interpreted as asymptotic precision without Monte Carlo uncertainty and observed-data correction.

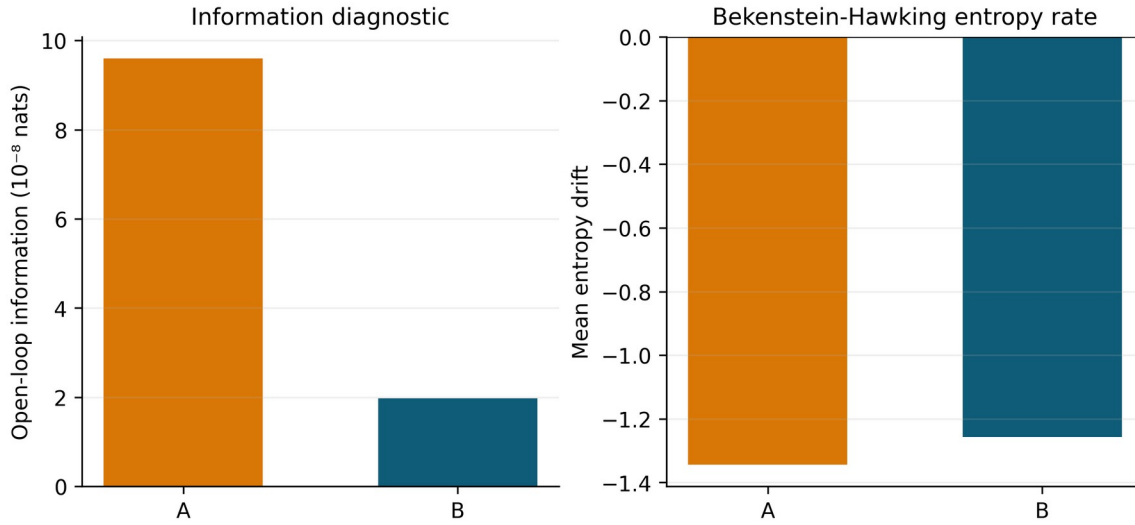
The requested diffusion carries much greater local curvature for $\tilde{\alpha}$ in this short-horizon complete-path experiment, whereas the packet process is slightly more informative for M_0 . This is not paradoxical: the Gaussian increment likelihood scores every time step for drift, while the sparse packet sample contained only two events in total on average across 16 paths. The result may reverse under a longer horizon, event-rich regime, common detector, or observed-data likelihood.

Information and Entropy Findings

The stored engine returned $I^A = 9.604098 \times 10^{-8}$ nats and $I^B = 1.977358 \times 10^{-8}$ nats, giving $\Delta I = -7.626740 \times 10^{-8}$ nats. The corresponding pathwise derivative of ΔI with respect to (M_0, γ, α) was approximately $(8.119 \times 10^{-8}, -4.802 \times 10^{-8}, 3.723 \times 10^{-10})$. Mean local entropy drifts were -1.344414 for A and -1.256689 for B.

Figure 5

Exploratory Information Diagnostic and Entropy Drift



Note. The left panel is not transfer entropy: both reduced predictors were open-loop leave-one-out estimates rather than filters adapted to each observation history.

The sign of ΔI is aligned with the masking hypothesis, but its inferential status is strictly exploratory. Three confounds remain: the requested diffusion is not moment matched, the effective detectors are not common, and $R = 16$ provides no stable uncertainty interval. The entropy comparison is better grounded because it follows the generator of each stopped process, yet it too depends on the cutoff and local-law specification.

Identifiability and Model Geometry

Identifiability failures are expected rather than exceptional. A short record identifies γ/M through counts, so γ is confounded with uncertain M_0 unless packet energies or direct mass calibration are retained. In the diffusion, α and M_0 share the lifetime combination M_0^3/α , while σ_0 can trade off against detector noise. Near m_s , large derivatives can create apparent local identifiability even when posterior mass follows a long global ridge.

Table 4

Primary Identifiability Risks and Controls

Risk	Mechanism	Required control
Structural zero	Inactive parameter within each paradigm	Common coordinates or separate vectors
Lifetime ridge	$\tau_0 \approx M_0^3/(3\alpha)$	Early mass calibration and intermediate telemetry
Count-mass confounding	Counts identify γ/M	Retain energy marks
Variance confounding	Process noise resembles detector noise	Repeated calibration and informative priors
Terminal stiffness	Derivatives diverge as M approaches zero	Positive cutoff and refinement studies
Unknown event count	Latent A path is trans-dimensional	Marginal, particle, RJ, or hybrid MCMC

Note. Controls should be pre-specified and evaluated through simulation-based calibration rather than selected after observing favorable residuals.

The event model also has trans-dimensional latent structure when emissions are not directly observed. The number of latent events changes across trajectories, so a naïve fixed-dimensional HMC parameterization is invalid. Options include exact marginalization in special cases, particle marginal methods, reversible-jump moves, or a hybrid scheme in which HMC updates continuous parameters conditional on an event configuration. Observed-data Fisher information should be estimated after integrating this missing structure.

MCMC Strategy, Falsification Plan, and Conclusions

HMC/NUTS Implementation

For Paradigm B, sample unconstrained parameters and standard-normal innovations rather than the mass path directly. Noncentering weakens funnel geometry, while a dense or block mass matrix can accommodate the M_0 - α ridge. Adaptation must be evaluated with divergences, energy diagnostics, effective sample sizes, and rank-normalized $\hat{R}$ (Hoffman & Gelman, 2014; Vehtari et al., 2021). For Paradigm A, conditionally continuous HMC updates should be embedded within a sampler that changes event topology.

Decisive Study

The confirmatory experiment should pre-register (a) a common detector K ; (b) an A-versus-B★ moment-matched primary contrast; (c) observation-history causal filters; (d) $\Delta I(t)$, integrated $\Delta I(T)$, and common-coordinate observed Fisher matrices; (e) Monte Carlo confidence or posterior intervals; and (f) a grid of horizons, cutoffs, time steps, detector resolutions, and greybody/mark laws. Simulation-based calibration should verify coverage before analyzing scientific runs (Talts et al., 2018).

Conclusion

The project now has a coherent mathematical experiment and a differentiable software foundation. Discrete emissions and continuous flow can be compared without conflating sample spaces by using stopped filtrations, a common detector, likelihood-ratio information functionals, and common Fisher coordinates. The validated baseline shows a smaller open-loop information diagnostic under the requested diffusion, but it does not yet establish information masking. The scientifically defensible conclusion is conditional: the hypothesis remains plausible and testable, and the next-stage controls are sufficiently explicit that both confirmation and falsification are possible. A robust nonnegative ΔI after matching and causal filtering would disprove the directional hypothesis; a stable negative residual across controls would support it.

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Appendix A

Python Code for All Figures and Data Tables

The following complete script reads the repository's validated JSON artifacts and reproduces Figures 1–5 and Tables 1–4.

```

"""Reproduce every graph and data table used in the research report.

The script reads the validated JSON artifacts produced by the repository's
PyTorch pipeline. It does not relabel the open-loop diagnostic as transfer
entropy. Figures are saved at publication resolution and tables as CSV files.
"""

from __future__ import annotations

import csv
import json
from pathlib import Path

import matplotlib.pyplot as plt
import numpy as np
from matplotlib.patches import FancyArrowPatch, FancyBboxPatch

ROOT = Path(__file__).resolve().parents[1]
ARTIFACTS = ROOT / "black-hole-information-dynamics" / "artifacts"
FIGURES = ROOT / "report_work" / "figures"
TABLES = ROOT / "report_work" / "tables"
FIGURES.mkdir(parents=True, exist_ok=True)
TABLES.mkdir(parents=True, exist_ok=True)

AMBER = "#D97706"
BLUE = "#0F5C78"
PURPLE = "#6D4C9C"
GRAY = "#56616F"
KAPPA = 1.0 / (8.0 * np.pi)

plt.rcParams.update({
    "font.family": "DejaVu Sans",
    "font.size": 10,
    "axes.titlesize": 11,
    "axes.labelsize": 10,
    "figure.dpi": 160,
    "savefig.dpi": 300,
    "axes.spines.top": False,
    "axes.spines.right": False,
})

def load_json(name: str) -> dict:
    with (ARTIFACTS / name).open(encoding="utf-8") as stream:
        return json.load(stream)

part0 = load_json("part0/model_validation.json")
part1 = load_json("part1/paradigm_a_summary.json")
part2 = load_json("part2/paradigm_b_summary.json")
part3 = load_json("part3/information_summary.json")

def save(fig: plt.Figure, name: str) -> None:
    fig.savefig(FIGURES / name, bbox_inches="tight", facecolor="white")
    plt.close(fig)

```

Appendix A (continued): Code, Part 2

```
def figure_architecture() -> None:
    fig, ax = plt.subplots(figsize=(8.0, 3.8))
    ax.set_xlim(0, 10)
    ax.set_ylim(0, 5)
    ax.axis("off")
    nodes = {
        "Latent mass\nM(t)": (0.5, 2.0, 1.7, 1.0, PURPLE),
        "A: marked packet\nnprocess": (3.0, 3.2, 2.0, 1.0, AMBER),
        "B: Itô mass\ndiffusion": (3.0, 0.8, 2.0, 1.0, BLUE),
        "Common detector\noperator K": (6.0, 2.0, 1.9, 1.0, GRAY),
        "Filtration-aware\ninformation\nengine": (8.25, 2.0, 1.65, 1.0, PURPLE),
    }
    for label, (x, y, w, h, color) in nodes.items():
        patch = FancyBboxPatch((x, y), w, h, boxstyle="round,pad=0.08",
                               edgecolor=color, facecolor="white", linewidth=1.8)
        ax.add_patch(patch)
        font_size = 8.8 if label.startswith("Filtration") else 10
        ax.text(x + w / 2, y + h / 2, label, ha="center", va="center",
               color=color, fontsize=font_size)
    arrows = [((2.2, 2.5), (3.0, 3.7)), ((2.2, 2.5), (3.0, 1.3)),
              ((5.0, 3.7), (6.0, 2.65)), ((5.0, 1.3), (6.0, 2.35)),
              ((7.9, 2.5), (8.25, 2.5))]
    for start, end in arrows:
        ax.add_patch(FancyArrowPatch(start, end, arrowstyle="-|>", mutation_scale=12,
                                     color="#7A838E", linewidth=1.2))
    ax.text(5.0, 4.7, "Controlled comparison of statistical experiments",
           ha="center", va="top", weight="bold", fontsize=12)
    save(fig, "figure_1_architecture.png")

def figure_scalings() -> None:
    config = part0["config"]
    gamma = config["theta"]["gamma"]
    alpha = config["theta"]["alpha"]
    sigma0 = config["sigma_base"]
    c2 = part0["matching"]["planck_second_moment_x"]
    mass = np.geomspace(config["m_floor"], config["theta"]["M0"], 300)
    curves = {
        "Packet intensity  $\gamma/M$ ": gamma / mass,
        "Drift magnitude  $\alpha/M^2$ ": alpha / mass**2,
        "Requested noise  $\sigma_0/M$ ": sigma0 / mass,
        "Matched noise  $\sqrt{(\gamma\kappa^2 c_2)M^{-3/2}}$ ": np.sqrt(gamma * KAPPA**2 * c2) / mass**1.5,
    }
    fig, ax = plt.subplots(figsize=(7.2, 4.3))
    colors = [AMBER, BLUE, GRAY, PURPLE]
    styles = ["-", "-", "-.-", "-."]
    for (label, values), color, style in zip(curves.items(), colors, styles):
        ax.loglog(mass, values, label=label, color=color, linestyle=style, linewidth=2)
    ax.axvline(config["m_floor"], color="#222222", linestyle=":", linewidth=1)
    ax.set_xlabel("Mass M (Planck units)")
    ax.set_ylabel("Local coefficient magnitude")
    ax.set_title("Singular local scaling near the stopping boundary")
    ax.grid(True, which="both", alpha=0.18)
    ax.legend(frameon=False, fontsize=8.5)
    save(fig, "figure_2_scalings.png")
```

Appendix A (continued): Code, Part 3

```

def figure_terminal_mass() -> None:
    labels = ["A: packets", "B: requested", "B*: matched"]
    means = [part1["summary"]["mean_final_mass"],
              part2["requested_B"]["mean_final_mass"],
              part2["moment_matched_B_star"]["mean_final_mass"]]
    sds = [part1["summary"]["final_mass_sd"],
           part2["requested_B"]["final_mass_sd"],
           part2["moment_matched_B_star"]["final_mass_sd"]]
    fig, ax = plt.subplots(figsize=(6.8, 3.8))
    x = np.arange(3)
    for xpos, mean, sd, color in zip(x, means, sds, [AMBER, BLUE, PURPLE]):
        ax.errorbar(xpos, mean, yerr=sd, fmt="o", markersize=8, capsize=6,
                    color=color, ecolor=color, linewidth=1.8)
    for xpos, mean in zip(x, means):
        ax.text(xpos, mean + 0.014, f"{mean:.6f}", ha="center", fontsize=8.5)
    ax.set_xticks(x, labels)
    ax.set_ylabel("Terminal mass at T = 0.2")
    ax.set_title("Exploratory terminal-mass summaries (mean  $\pm$  1 SD, R = 16)")
    ax.set_ylim(min(np.array(means) - np.array(sds)) - 0.012,
                max(np.array(means) + np.array(sds)) + 0.028)
    ax.grid(axis="y", alpha=0.2)
    save(fig, "figure_3_terminal_mass.png")

def figure_fisher() -> None:
    result = part3["result"]
    matrices = [np.array(result["fisher_a_common_M0_alpha_eff"]),
                 np.array(result["fisher_b_common_M0_alpha_eff"]),
                 np.array(result["delta_fisher_common_B_minus_A"])]
    titles = ["A: packet process", "B: requested diffusion", "Residual B - A"]
    vmax = max(np.max(np.abs(matrix)) for matrix in matrices)
    fig, axes = plt.subplots(1, 3, figsize=(8.2, 3.2), constrained_layout=True)
    image = None
    for ax, matrix, title in zip(axes, matrices, titles):
        image = ax.imshow(matrix, cmap="PuOr", vmin=-vmax, vmax=vmax)
        ax.set_xticks([0, 1], ["M0", "α̃"])
        ax.set_yticks([0, 1], ["M0", "α̃"])
        ax.set_title(title)
        for i in range(2):
            for j in range(2):
                ax.text(j, i, f"{matrix[i, j]:.3f}", ha="center", va="center",
                        color="white" if abs(matrix[i, j]) > 0.35 * vmax else "black",
                        fontsize=8.5, weight="bold")
    fig.colorbar(image, ax=axes, shrink=0.75, label="Fisher information")
    save(fig, "figure_4_fisher.png")

def figure_information_entropy() -> None:
    result = part3["result"]
    fig, axes = plt.subplots(1, 2, figsize=(7.6, 3.5), constrained_layout=True)
    info = np.array([result["information_a_nats"], result["information_b_nats"]]) * 1e8
    axes[0].bar(["A", "B"], info, color=[AMBER, BLUE], width=0.58)
    axes[0].set_ylabel("Open-loop information (10-8 nats)")
    axes[0].set_title("Information diagnostic")
    axes[0].grid(axis="y", alpha=0.2)
    entropy = [result["mean_entropy_rate_a"], result["mean_entropy_rate_b"]]
    axes[1].bar(["A", "B"], entropy, color=[AMBER, BLUE], width=0.58)
    axes[1].axhline(0, color="black", linewidth=0.8)

```

Appendix A (continued): Code, Part 4

```

axes[1].set_ylabel("Mean entropy drift")
axes[1].set_title("Bekenstein-Hawking entropy rate")
axes[1].grid(axis="y", alpha=0.2)
save(fig, "figure_5_information_entropy.png")

def write_csv(name: str, header: list[str], rows: list[list[object]]) -> None:
    with (TABLES / name).open("w", newline="", encoding="utf-8") as stream:
        writer = csv.writer(stream)
        writer.writerow(header)
        writer.writerows(rows)

def write_tables() -> None:
    config = part0["config"]
    write_csv("table_1_model_comparison.csv", ["Feature", "Paradigm A", "Paradigm B"], [
        ["Latent law", "Stopped marked point process", "Stopped Itô diffusion"],
        ["Mass change", "Discrete energy jumps", "Continuous Euler-Maruyama increments"],
        ["Local driver", " $\lambda(M)=\gamma/M$ ", " $b(M)=-\alpha/M^2$ ;  $s(M)=\sigma_0/M$ "],
        ["Natural data", "Event times and energy marks", "Continuous noisy telemetry"],
        ["Inactive parameter", " $\alpha$ ", " $\gamma$ "],
    ])
    write_csv("table_2_configuration.csv", ["Quantity", "Symbol", "Baseline"], [
        ["Initial mass", " $M_0$ ", config["theta"]["M0"]],
        ["Packet-rate scale", " $\gamma$ ", config["theta"]["gamma"]],
        ["Diffusion drift", " $\alpha$ ", config["theta"]["alpha"]],
        ["Time horizon", " $T$ ", config["horizon"]],
        ["Euler step", " $\Delta t$ ", config["dt"]],
        ["Stopping mass", " $m_s$ ", config["m_floor"]],
        ["Diffusion scale", " $\sigma_0$ ", config["sigma_base"]],
        ["Replicates", "R", part3["provenance"]["replicates"]],
    ])
    result = part3["result"]
    write_csv("table_3_baseline_findings.csv", ["Estimand", "Value", "Status"], [
        ["IA", " $9.6041 \times 10^{-8}$ ", "Open-loop diagnostic"],
        ["IB", " $1.9774 \times 10^{-8}$ ", "Open-loop diagnostic"],
        [" $\Delta I = I^B - I^A$ ", " $-7.6267 \times 10^{-8}$ ", "Exploratory"],
        ["Mean entropy drift A", f"{result['mean_entropy_rate_a']:.6f}", "Validated calculation"],
        ["Mean entropy drift B", f"{result['mean_entropy_rate_b']:.6f}", "Validated calculation"],
        ["Unit tests", 6, "All passed"],
    ])
    write_csv("table_4_identifiability.csv", ["Risk", "Mechanism", "Required control"], [
        ["Structural zero", "Inactive parameter within each paradigm", "Common coordinates or separate
vectors"],
        ["Lifetime ridge", " $\tau_0 \approx M_0^3/(3\alpha)$ ", "Early mass calibration and intermediate telemetry"],
        ["Count-mass confounding", "Counts identify  $\gamma/M$ ", "Retain energy marks"],
        ["Variance confounding", "Process noise resembles detector noise", "Repeated calibration and
informative priors"],
        ["Terminal stiffness", "Derivatives diverge as M approaches zero", "Positive cutoff and refinement
studies"],
        ["Unknown event count", "Latent A path is trans-dimensional", "Marginal, particle, RJ, or hybrid
MCMC"],
    ])

if __name__ == "__main__":
    figure_architecture()
    figure_scalings()
    figure_terminal_mass()
    figure_fisher()
    figure_information_entropy()
    write_tables()
    print(f"Wrote 5 figures to {FIGURES} and 4 tables to {TABLES}.")

```